PROJECT SCOPE & TECHNICAL SPECIFICATION

Windows Workforce Telemetry & Task Management System

PREPARED FOR:	SCOPE TIER:	TARGET OS:	ARCHITECTURE:
Anti-Gravity Tech Team	Phase 1 (MVP) Specification	Windows 10 / 11 (64-bit)	Native C# Agent + Cloud API

Executive Summary: The goal of Phase 1 is to deploy a lightweight, native Windows agent combined with a central web management backend. The system must reliably automate employee attendance/activity telemetry (logon/logoff, idle time, active applications) and provide full task lifecycle management (assignment, status progression, and work deliverable uploads).

1. Scope Matrix: Must-Have vs. Nice-to-Have (Phase 1)

Feature Module	Classification	Technical Scope & Behavior	Success Criteria
Session Event Logging	MUST-HAVE	Capture OS Logon, Logoff, Workstation Lock, and Unlock via `SystemEvents.SessionSwitch` and Security Event Logs (4624/4634/4800/4801).	Zero missed logon/ lock timestamps; accurate down to the second.
Active Window Tracking	MUST-HAVE	Win32 event hook on `EVENT_SYSTEM_FOREGROUND`. Capture process name, window title, and active focused duration.	Client CPU < 1%; exact time breakdown by active application.
Idle Time Detection	MUST-HAVE	Query `GetLastInputInfo()`. When idle duration > 300s (5 mins), split telemetry into active vs. idle hours.	Distinguishes active desk work from extended breaks.
Task Assignment & Feed	MUST-HAVE	Desktop UI showing assigned tasks. Employee toggles state: *Assigned → In Progress → Completed*.	Employee sees work instantly; updates sync upstream.
Task Time Linking	MUST-HAVE	When employee clicks "Start Task", link ongoing app telemetry to that specific `task_id`.	Managers see actual hours and apps used per ticket.
Deliverable File Upload	MUST-HAVE	Drag-and-drop file upload against a task (ZIP, PDF, DOCX, etc.) via Presigned S3 URLs (capped at 50MB).	Completed files stored in cloud object storage securely.
Offline Telemetry Buffer	MUST-HAVE	Local encrypted SQLite database to store telemetry logs if network drops; auto-syncs on reconnect.	No log loss during Internet outages or remote work travel.
Realtime Toast Alerts	NICE-TO-HAVE	SignalR-powered native Windows Toast notifications when a new task is assigned to the employee.	Immediate alert without manual UI refresh.
Periodic Screenshots	NICE-TO-HAVE	Screen capture every 10–15 minutes with optional privacy blur mode for password/sensitive fields.	Optional visual proof-of-work. (Can defer to Phase 2).
	NICE-TO-HAVE		

Feature Module	Classification	Technical Scope & Behavior	Success Criteria
Productivity Categorization		Rule-based tagging of apps as Productive (VS Code, Excel), Communication (Slack), or Neutral.	Automated productivity index score.
USB / File Transfer DLP	FUTURE (PHASE 2)	Detection and logging of peripheral storage insertion or bulk file copy operations.	Enterprise data loss prevention compliance.

2. Core Technical Architecture & Boundaries

A. Desktop Client Agent (.NET 8 / C# WPF or WinUI 3)

- **Process Footprint:** Must consume < 60 MB RAM and < 1% CPU utilization during continuous monitoring.
- **Background Worker:** Runs as a Windows Service or tray-docked background service with restart resilience.
- **Telemetry Synchronization:** Batches window logs and idle events locally, sending an HTTP payload every 60 seconds to `POST /api/v1/telemetry/batch`.

B. Central Backend & Cloud Infrastructure

- **API Stack:** ASP.NET Core Web API or Node.js/NestJS exposing documented REST endpoints.
- **Database:** PostgreSQL 16+ storing Relational Users, Tasks, Sessions, and Aggregated Activity.
- **File Storage:** S3-compatible Object Storage (AWS S3, MinIO, or Cloudflare R2) using Presigned direct-upload URLs.
- **Manager Web Portal:** React/Next.js interface for managing task distribution, viewing employee logs, and reviewing attachments.

3. Phase 1 Delivery Milestones (Target: 8 Weeks)

1. **Sprint 1 (Weeks 1–2):** Database schema migration, Authentication (JWT), Task CRUD API, S3 Presigned URL integration.
2. **Sprint 2 (Weeks 3–4):** C# Win32 telemetry agent (active window hook, idle detection, session switch events, local SQLite cache).
3. **Sprint 3 (Weeks 5–6):** Desktop UI dashboard (task lists, timer start/stop, drag-and-drop file upload client, offline sync).
4. **Sprint 4 (Weeks 7–8):** Manager web dashboard, reporting charts (daily hours, app breakdown), MSI installer package & QA testing.

4. Explicit Phase 1 Out-of-Scope Items

- Keystroke logging / keylogger modules (intentionally excluded to ensure security and privacy compliance).
- Remote desktop control / live screen streaming (bandwidth and infrastructure heavy).
- Automated biometric or webcam facial recognition.